

Yoshi Sato

Incoming PhD Student | Systems Security & Confidential Computing

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EDUCATION

YALE UNIVERSITY

Doctor of Philosophy in Computer Science

New Haven, CT

Sept 2026 - May 2031

WASEDA UNIVERSITY

Bachelor of Science in Computer Science and Communications Engineering

Tokyo, Japan

Oct 2022 - Jul 2026

Research Fields: Systems Security, Confidential Computing, Cloud Infrastructure, Secure Computer Architecture

PUBLICATIONS

[Poster Paper] **Y. Sato**, H. Uranami, A. Saiki, K. Kimura. "Towards GPU Passthrough in Intel TDX: Design Challenges and Early Baselines." In *Proceedings of the IEEE 23rd International Conference on Dependable, Autonomic and Secure Computing (DASC)*, 2025. (Best Poster Award)

[Preprint] M. Barbaraci, S. Ma, H. Malvai, M. Moullem, S. Ren, **Y. Sato**, S. Yang, F. Zhang. "DeadDrop: Responsible Disclosure of Smart Contract Bugs." Submitted to *Proceedings on Privacy Enhancing Technologies (PoPETS)*, 2026.
(Author order alphabetical)

RESEARCH EXPERIENCE

Student Researcher, Kasahara-Kimura Lab, Waseda University

Apr 2025 - Present

Advisor: Prof. Keiji Kimura & Prof. Hironori Kasahara

- Writing *Demystifying Coconut-SVSM: An Architectural and Empirical Study of an AMD SEV-SNP Paravisor*, an ongoing systems study quantifying SVSM-induced overheads on CVM boot (**45% cold-start latency increase over non-SVSM SEV-SNP**), runtime execution, and trusted service calls (e.g., vTPM and attestation).
- Conducting cross-layer low-level systems analysis of Coconut-SVSM, including x86-64 boot code, privileged CPU state, SEV-SNP/VMPL mechanisms, IGVM-based launch structure, and Linux guest/SVSM call paths to explain observed performance bottlenecks.
- Initiated an independent research project on GPU passthrough in Intel TDX, culminating in a paper accepted to IEEE DASC 2025 that won the Best Poster Award.
- Identified that while AMD's PVALIDATE incurs zero VM exits, Intel TDX's TDG.MEM.PAGE.ACCEPT triggers mandatory VM exits, creating a structural latency bottleneck.
- Quantified memory overheads, establishing a baseline revealing that pageable memory throughput degrades by ~53% compared to pinned memory in confidential environments.

Research Intern, Decentralized Systems Group, Yale University

Jul - Oct 2025

Advisor: Prof. Fan Zhang

- Architected the core cryptographic primitive for DeadDrop using Oblivious Message Retrieval (OMR) to produce the first functional prototype for oblivious bug reporting.
- Identified a critical "database jumbling" vulnerability where implementation deviated from the theoretical binding of payloads and clues; led discussions to redefine the system's threat model.
- Designed and executed performance evaluations to quantify clue processing, achieving a throughput of 32,768 payloads in 108s (single-core) and scaling to 524,288 payloads in 110s (16-core).

- Parallelized the precomputation phase of OMR to **reduce initialization latency by 65%**, significantly improving the prototype's practicality for real-world deployment.
- Leading a Systematization of Knowledge (SoK) project on TEEs with researchers from Yale and UIUC to define a new taxonomy and evaluation framework for confidential containers.

Research Intern, IJ Research Laboratory

Mar 2025 - Present

Advisor: Dr. Pierre-Louis Aublin

- Collaborated with the LSDS Group at Imperial College London on Serverless Confidential Containers (SC2) to analyze performance overheads in secure serverless computing.
- Transformed a research prototype into a fully reproducible testbed, enabling the first performance evaluation of the SC2 system on Intel TDX hardware.
- Engineered custom network proxy support for the Knative control plane to resolve restricted-network deployment failures and analyzed CVM cold-start paths with Ftrace to identify dominant latency sources.

PROFESSIONAL SERVICE

Artifact Evaluation Committee (AEC) Member

Feb 2026

The 21st European Conference on Computer Systems (EuroSys 2026)

Student Volunteer

Jun 2025

The 52nd Annual International Symposium on Computer Architecture (ISCA 2025)

AWARDS

Best Poster Award, IEEE DASC 2025

PROFESSIONAL EXPERIENCE

IoT Penetration Testing Intern, FFRI Security

Jan - Feb 2025

- Conducted static and dynamic binary analysis on IoT firmware and malware executables using Ghidra and GDB, identifying critical vulnerabilities in legacy software components.
- Executed penetration tests on simulated IoT environments to simulate real-world attack vectors against embedded devices, using Nmap, Metasploit, and Wireshark.

LEADERSHIP

Secretary, IEEE-HKN, Mu Tau Chapter

Jul 2025 - Present

Nominated by 2018 IEEE Computer Society President

Co-founder, Kuma Lab

Apr 2024 - Present

- Founded a student research community of ~30 members to foster peer learning in AI and robotics.
- Organized a technical workshop hosted at Google Japan HQ for 25 attendees to bridge the gap between advanced research and non-STEM audiences.

SKILLS

- TEEs: Intel TDX, AMD SEV-SNP, VMPLs, Paravisor, vTPM
- Systems: QEMU, Kubernetes, Knative, Docker, Linux/KVM, OVMF/EDK2
- Applied Cryptography: Microsoft SEAL, PALISADE
- Programming Languages: Bash, Rust, C++, Python